7.7 Hyperbolic functions

Formed by taking combinations of e^x and e^{-x}.

$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2}$$

Cosh x (Even) — ***Hyperbolic cosine***

$\sinh x$ (Odd) — ***Hyperbolic sine***

Remark: f is even if $f(-x) = f(x)$ $\forall\, x \in D_f$.
f is odd if $f(-x) = -f(x)$ $\forall\, x \in D_f$.

Definition:

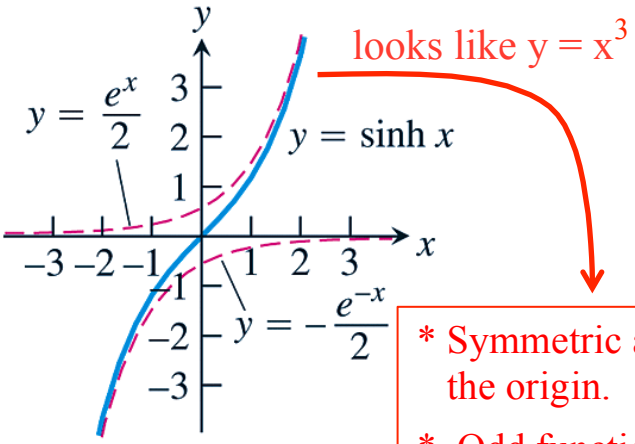
$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}, \qquad \operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}},$$

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}.$$

TABLE 7.5 The six basic hyperbolic functions

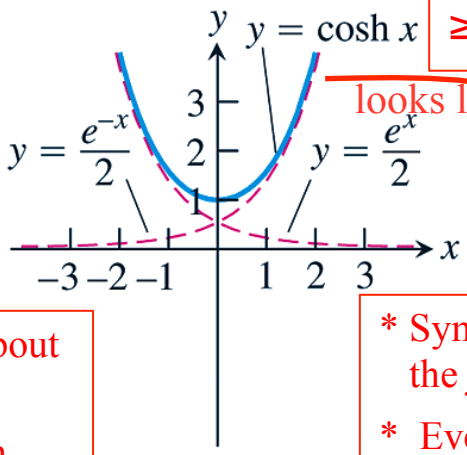


(a)

looks like $y = x^3$

* Symmetric about the origin.

* Odd function.



(b)

looks like $y = x^2 + 1$

* Symmetric about the *y*-axis.

* Even function.

Hyperbolic sine:

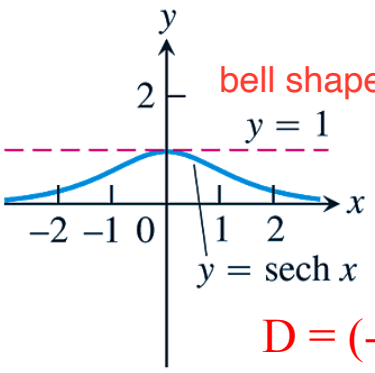
$$\sinh x = \frac{e^x - e^{-x}}{2}$$

D = (–∞, ∞)
R = (–∞, ∞)

Hyperbolic cosine:

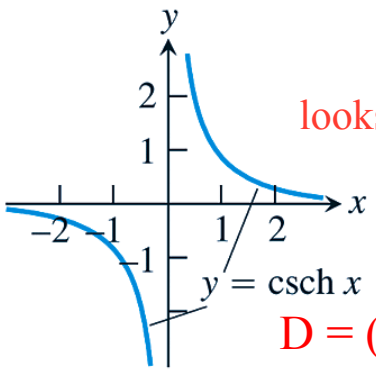
$$\cosh x = \frac{e^x + e^{-x}}{2}$$

D = (–∞, ∞)
R = [1, ∞)



(d)

D = (–∞, ∞)
R = (0, 1]



(e)

looks like $y = \frac{1}{x}$

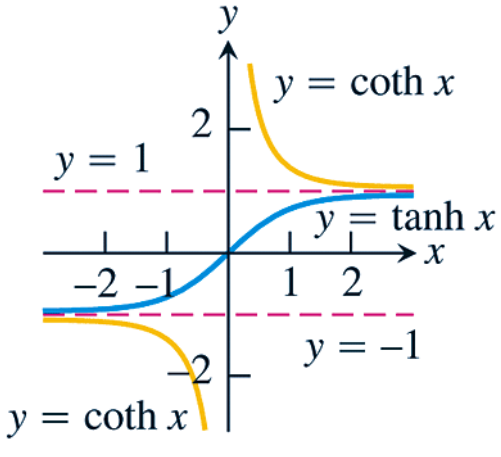
D = (–∞, 0) ∪ (0, ∞)
R = (–∞, 0) ∪ (0, ∞)

Hyperbolic secant:

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

Hyperbolic cosecant:

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$



(c)

Hyperbolic tangent:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

D = (–∞,∞)
R = (–1, 1)

Hyperbolic cotangent:

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

D = (–∞, 0) ∪ (0, ∞)
R = (–∞,–1) ∪ (1, ∞)

Identities for Hyperbolic functions (Table 7.6)

Hyperbolic functions	Trigonometric Functions
$\cosh^2 x - \sinh^2 x = 1$	$\cos^2 x + \sin^2 x = 1$
$\sinh 2x = 2\sinh x \cosh x$	$\sin 2x = 2\sin x \cos x$
$\cosh 2x = \cosh^2 x + \sinh^2 x$	$\cos 2x = \cos^2 x - \sin^2 x$
$\cosh^2 x = \frac{\cosh 2x + 1}{2}$	$\cos^2 x = \frac{\cos 2x + 1}{2}$
$\sinh^2 x = \frac{\cosh 2x - 1}{2}$	$\sin^2 x = \frac{1 - \cos 2x}{2}$
$\tanh^2 x = 1 - \operatorname{sech}^2 x$	$\tan^2 x = \sec^2 x - 1$
$\coth^2 x = 1 + \operatorname{csch}^2 x$	$\cot^2 x = \csc^2 x - 1$

Example 1: $2\sinh x \cosh x = 2\left(\frac{e^x - e^{-x}}{2}\right)\left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^{2x} - e^{2x}}{2} = \sinh 2x.$

Example 2: Show that (1) $\cosh^2 x - \sinh^2 x = 1$

(2) $\cosh 2x = \cosh^2 x + \sinh^2 x$

(3) $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$

Solution:

(1) $\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$

$= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4} = \frac{4}{4} = 1$

(2) $\cosh^2 x + \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 + \left(\frac{e^x - e^{-x}}{2}\right)^2 = \frac{e^{2x} + 2 + e^{-2x}}{4} + \frac{e^{2x} - 2 + e^{-2x}}{4}$

$= \frac{2e^{2x} + 2e^{-2x}}{4} = \frac{1}{2}\left(\frac{e^x + e^{-x}}{2}\right)^2 = \cosh 2x$

(3) $\cosh x \cosh y + \sinh x \sinh y = \left(\frac{e^x + e^{-x}}{2}\right)\left(\frac{e^y + e^{-y}}{2}\right) + \left(\frac{e^x - e^{-x}}{2}\right)\left(\frac{e^y - e^{-y}}{2}\right)$

$= \frac{e^{x+y} + e^{x-y} + e^{y-x} + e^{-(x+y)}}{4} + \frac{e^{x+y} - e^{x-y} - e^{y-x} + e^{-(x+y)}}{4} = \frac{e^{x+y} + e^{-(x+y)}}{2}$

$= \cosh(x + y)$

Derivative of Hyperbolic functions

TABLE 7.7 Derivatives of hyperbolic functions

$$\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$$

$$\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$$

$$\frac{d}{dx}(\tanh u) = \operatorname{sech}^2 u \frac{du}{dx}$$

$$\frac{d}{dx}(\coth u) = -\operatorname{csch}^2 u \frac{du}{dx}$$

$$\frac{d}{dx}(\operatorname{sech} u) = -\operatorname{sech} u \tanh u \frac{du}{dx}$$

$$\frac{d}{dx}(\operatorname{csch} u) = -\operatorname{csch} u \coth u \frac{du}{dx}$$

Similar to trigonometric function EXCEPT cosh x, sech x } differences in sign

The derivative formulas are derived from the derivative of e^u:

$$\frac{d}{dx}(\sinh u) = \frac{d}{dx}\left(\frac{e^u - e^{-u}}{2}\right) \quad \text{Definition of sinh } u$$

$$= \frac{e^u\, du/dx + e^{-u}\, du/dx}{2} \quad \text{Derivative of } e^u$$

$$= \cosh u \frac{du}{dx} \quad \text{Definition of cosh } u$$

This gives the first derivative formula. The calculation

$$\frac{d}{dx}(\operatorname{csch} u) = \frac{d}{dx}\left(\frac{1}{\sinh u}\right) \quad \text{Definition of csch } u$$

$$= -\frac{\cosh u}{\sinh^2 u}\frac{du}{dx} \quad \text{Quotient Rule}$$

$$= -\frac{1}{\sinh u}\frac{\cosh u}{\sinh u}\frac{du}{dx} \quad \text{Rearrange terms.}$$

$$= -\operatorname{csch} u \coth u \frac{du}{dx} \quad \text{Definitions of csch } u \text{ and coth } u$$

The other formulas in Table 7.7 are obtained similarly.

Derivatives of Trigonometric, Inverse Trigonometric, Hyperbolic and Inverse Hyperbolic functions

Table 1

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

Table 7.7

Derivatives of Hyperbolic Functions

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$$

$$\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$$

Table 7.3

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

Table 7.10

Derivatives of Inverse Hyperbolic Functions

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}, \quad x > 1$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}, \quad |x| < 1$$

$$\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}, \quad |x| > 1$$

$$\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}, \quad 0 < x < 1$$

$$\frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{1+x^2}}, \quad x \neq 0$$

Example 3: (1) Find $\frac{d}{dx}\cosh\left(\ln\sqrt{1+x^2}\right)$.

(2) Show that (a) $\frac{d}{dx}e^{2x}\cosh 2x = 2e^{2x}$

(b) $\frac{d}{dx}\left(x^2-1\right)\operatorname{csch}\left(\ln x\right)=2$.

Solution:

(1) $\frac{d}{dx}\cosh\left(\ln\sqrt{1+x^2}\right)=\sinh\left(\ln\sqrt{1+x^2}\right)\frac{d}{dx}\left(\frac{1}{2}\ln(1+x^2)\right)=\frac{x\sinh\left(\ln(1+x^2)\right)}{1+x^2}$.

(2) (a) $\frac{d}{dx}e^{2x}\cosh 2x=\frac{d}{dx}e^{2x}\left(\frac{e^{2x}+e^{-2x}}{2}\right)=\frac{1}{2}\frac{d}{dx}\left(e^{4x}+1\right)=2e^{4x}$.

(b) $\left(x^2-1\right)\operatorname{csch}(\ln x)=\left(x^2-1\right)\left(\frac{2}{e^{\ln x}-e^{-\ln x}}\right)=\left(x^2-1\right)\left(\frac{2}{x-\frac{1}{x}}\right)=\left(x^2-1\right)\left(\frac{2x}{x^2-1}\right)=2x$.

$\therefore \quad \frac{d}{dx}\left(x^2-1\right)\operatorname{csch}\left(\ln x\right)=2$.

Inverse Hyperbolic Functions

The inverses of the six basic hyperbolic functions are used in integration.

Table 7.8

Function	Domain	Range	Graph
$\sinh^{-1} x$	$(-\infty,\infty)$	$(-\infty,\infty)$	Figure 7.32 (a)
$\cosh^{-1} x$	$[1, \infty)$	$[0, \infty)$	Figure 7.32 (b)
$\tanh^{-1} x$	$(-1, 1)$	$(-\infty,\infty)$	Figure 7.33 (a)
$\coth^{-1} x$	$(-\infty, -1) \cup (1, \infty)$	$(-\infty,0) \cup (0, \infty)$	Figure 7.33 (b)
$\operatorname{sech}^{-1} x$	$(0, 1]$	$[0, \infty)$	Figure 7.32 (c)
$\operatorname{csch}^{-1} x$	$(-\infty,0) \cup (0, \infty)$	$(-\infty,0) \cup (0, \infty)$	Figure 7.33 (c)

Remark:

1. $y = \sinh^{-1} x$ is the number in $(-\infty,\infty)$ for which $\sinh y = x$.
   $y = \cosh^{-1} x$ is the number in $[0, \infty)$ for which $\cosh y = x$.
   $y = \tanh^{-1} x$ is the number in $(-\infty,\infty)$for which $\tanh y = x$.

2. Take note that $\sinh^{-1}x \neq \frac{1}{\sinh x}$.

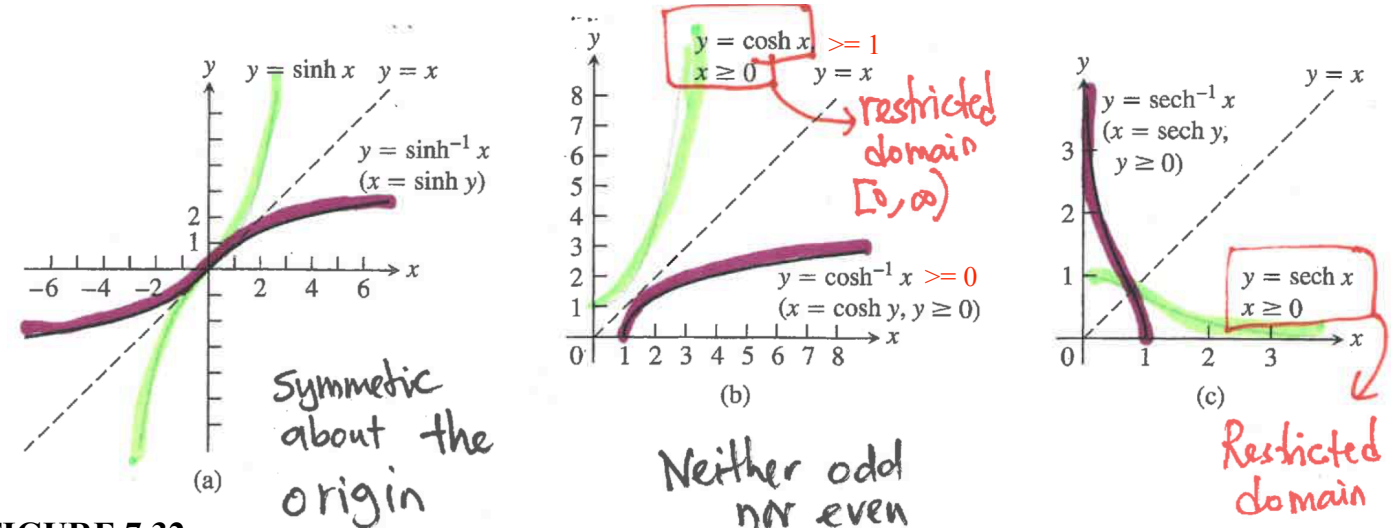


FIGURE 7.32 The graphs of the inverse hyperbolic sine, cosine, and secant of x. Notice the symmetries about the line $y = x$.

odd

$\sinh^{-1}(-x) = -\sinh^{-1} x$.

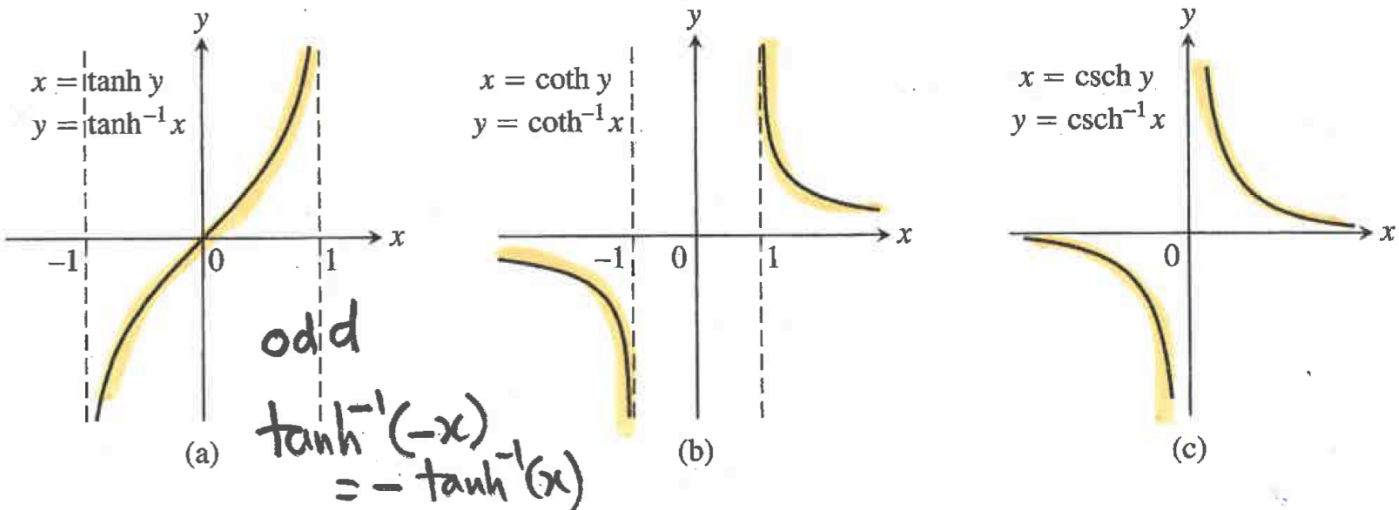


FIGURE 7.33 The graphs of the inverse hyperbolic tangent, cotangent, and cosecant of x.

Identities for Inverse Hyperbolic Functions

We use the identities in Table 7.9 to calculate the values of $\text{sech}^{-1} x$, $\text{csch}^{-1} x$, and $\coth^{-1} x$ on calculators that give only $\cosh^{-1} x$, $\sinh^{-1} x$, and $\tanh^{-1} x$.

Table 7.9:

(1) $\text{sech}^{-1} x = \cosh^{-1} \dfrac{1}{x}$, $x \in (0, 1]$.

(2) $\text{csch}^{-1} x = \sinh^{-1} \dfrac{1}{x}$, $x \in (-\infty, 0) \cup (0, \infty)$.

(3) $\coth^{-1} x = \tanh^{-1} \dfrac{1}{x}$, $(-\infty, -1) \cup (1, \infty)$.

Proof (1):

$$\text{sech}\left(\cosh^{-1}\left(\frac{1}{x}\right)\right) = \frac{1}{\cosh\left(\cosh^{-1}\left(\frac{1}{x}\right)\right)} = \frac{1}{\left(\frac{1}{x}\right)} = x.$$

$$\therefore \quad \cosh^{-1}\left(\frac{1}{x}\right) = \text{sech}\, x.$$

Derivative of Inverse Hyperbolic functions (Table 7.10)

TABLE 7.10 Derivatives of inverse hyperbolic functions

$$\frac{d(\sinh^{-1} u)}{dx} = \frac{1}{\sqrt{1 + u^2}} \frac{du}{dx}$$

$$\frac{d(\cosh^{-1} u)}{dx} = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx}, \qquad u > 1$$

$$\frac{d(\tanh^{-1} u)}{dx} = \frac{1}{1 - u^2} \frac{du}{dx}, \qquad |u| < 1$$

$$\frac{d(\coth^{-1} u)}{dx} = \frac{1}{1 - u^2} \frac{du}{dx}, \qquad |u| > 1$$

$$\frac{d(\text{sech}^{-1} u)}{dx} = -\frac{1}{u\sqrt{1 - u^2}} \frac{du}{dx}, \qquad 0 < u < 1$$

$$\frac{d(\text{csch}^{-1} u)}{dx} = -\frac{1}{|u|\sqrt{1 + u^2}} \frac{du}{dx}, \qquad u \neq 0$$

Example 4: Show that if u is a differentiable function of x whose values are greater than 1, then

$u > 1$

$$\frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}}\frac{du}{dx}.$$

Solution First we find the derivative of $y = \cosh^{-1} x$ for $x > 1$ by applying Theorem 1 of Section 7.1 with $f(x) = \cosh x$ and $f^{-1}(x) = \cosh^{-1} x$. Theorem 1 can be applied because the derivative of $\cosh x$ is positive when $x > 0$.

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \qquad \text{Theorem 1}$$

$$= \frac{1}{\sinh(\cosh^{-1} x)} \qquad f'(u) = \sinh u$$

change sinh u to cosh u by using $\cosh^2 u - \sinh^2 u = 1$

$$= \frac{1}{+\sqrt{\cosh^2(\cosh^{-1} x) - 1}} \qquad \cosh^2 u - \sinh^2 u = 1,\ \sinh u = +\sqrt{\cosh^2 u - 1}$$

+ only because $u = \cosh^{-1} x > 1$

$$= \frac{1}{\sqrt{x^2 - 1}} \qquad \cosh(\cosh^{-1} x) = x$$

The Chain Rule gives the final result:

$$\frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}}\frac{du}{dx}.$$

■

The above result also can be obtained by using implicit differentiation.

Example 5: Show that $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}, \quad |x| > 1$.

Solution: Let $y = \coth^{-1} x, \quad |x| > 1$. Then $\coth y = x$.

$$\frac{d}{dx}(\coth y) = \frac{dx}{dx}$$

$$\Rightarrow \quad -\operatorname{csch}^2 y \frac{dy}{dx} = 1$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{-\operatorname{csch}^2 y} = \frac{1}{1 - \coth^2 y} = \frac{1}{1-x^2}.$$

Example 6:

$$\frac{d}{dx}(\tanh^{-1}(\sin 2x)) = \frac{1}{1-(\sin 2x)^2}\frac{d}{dx}(\sin 2x) = \frac{2\cos 2x}{\cos^2 2x} = 2\sec 2x.$$